

MTH 446: Lecture # 19

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Lecture Span

1. Riemann Integration

Integration

$$\int_a^b w dt = \int_a^b u + i \int_a^b v \quad (1)$$

Therefore,

$$\operatorname{Re} \int w dt = \int \operatorname{Re} w \quad (2)$$

$$\operatorname{Im} \int w dt = \int \operatorname{Im} w \quad (3)$$

Consider a vector function in $\mathbb{R}^2$, then

$$\int_a^b f = \left(\int_a^b x, \int_a^b y \right) \quad (4)$$

Three properties:

1. $\int_a^b (f + g) = \int f + \int g$
2. $\operatorname{Re} \int w dt = \int \operatorname{Re} w$
3. $\operatorname{Im} \int w dt = \int \operatorname{Im} w$
4. $\int z_0 f = z_0 \int f$

We consider piecewise continuous functions. A function h is piecewise continuous if h is continuous at every point of $[a, b]$ except for a finite set of points $\{x_0, \dots, x_n\}$.

Here, $x_0 = a$ and $x_n = b$ WLOG. h also has a one-sided limit. Suppose $x_j \in \{a, b\}$, then the left and right limit of h as $x \rightarrow x_j$ must exist.

It is known that every piecewise continuous function is Riemann integrable. The fundamental theorem of calculus:

$$\int_a^b w' = w(b) - w(a) \quad (5)$$

This is true even for complex functions. Now, let w be piecewise continuous. Then, $|w|$ is also piecewise continuous.

Theorem 0.1. *Let $w : [a, b] \rightarrow \mathbb{C}$ be a piecewise continuous function. Then the following inequality holds:*

$$\left| \int_a^b w dt \right| \leq \int_a^b |w| dt \quad (6)$$

Proof. If $\int_a^b w dt = 0$, then we are done. Next, assume $\int_a^b w dt \neq 0$. Then,

$$\int_a^b w dt = r_0 \exp(i\theta_0) \quad (7)$$

$$\left| \int_a^b w dt \right| = r_0 \quad (8)$$

$$= \operatorname{Re} \left(\int_a^b \exp(-i\theta_0) w dt \right) \quad (9)$$

$$= \left| \int_a^b \operatorname{Re}(\exp(-i\theta_0) w) dt \right| \quad (10)$$

$$\leq \int_a^b |\operatorname{Re}(\exp(-i\theta_0) w)| dt \quad (11)$$

$$\left| \int_a^b w dt \right| = r_0 \leq \int_a^b |\operatorname{Re}(\exp(-i\theta_0) w)| dt \leq \int_a^b |\exp(-i\theta_0) w| dt = \int_a^b |w| dt \quad (12)$$

□

Example

Let

$$P_n(x) = \frac{1}{\pi} \int_0^\pi (x + i\sqrt{1-x^2} \cos \theta)^n d\theta \quad (13)$$

Then prove that $|P_n| \leq 1$. Then

$$|P_n| \leq \frac{1}{\pi} \int_0^\pi |x + i\sqrt{1-x^2} \cos \theta|^n d\theta \quad (14)$$

Moreover,

$$|x + i\sqrt{1-x^2} \cos \theta|^2 = x^2 + (1-x^2) \cos^2 \theta \leq x^2 + 1 - x^2 = 1 \quad (15)$$

Therefore,

$$|x + i\sqrt{1-x^2} \cos \theta|^n \leq 1 \quad (16)$$

Thus,

$$|P_n| \leq \frac{1}{\pi} \int_0^\pi d\theta = 1 \quad (17)$$

Definition 0.2. A Curve is a continuous function $\gamma : [a, b] \rightarrow \mathbb{C}$

Definition 0.3. A curve is closed if $\gamma(a) = \gamma(b)$

Definition 0.4. A curve γ is smooth if

1. $\gamma \in C^1$
2. $\gamma' \neq 0$ for all t .

Example, $\gamma = (t^2, t^3)$ is not smooth.

Definition 0.5. A curve g is called a simple curve if $t_1, t_2 \in [a, b]$ and $t_1 \neq t_2$, then $\gamma(t_1) \neq \gamma(t_2)$.

You can have a simple, closed curve. The arc length of a curve is

$$s(\gamma) = \int_a^b \sqrt{x'^2 + y'^2} dt \quad (18)$$

Note that $|\gamma'| = \sqrt{x'^2 + y'^2}$, and therefore, in complex notation, we have that:

$$s(\gamma) = \int_a^b |g'| dt \quad (19)$$